

**Diego Garza**  
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## EDUCATION

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| The University of California at Santa Cruz                             | Santa Cruz, CA      |
| PhD in Astronomy & Astrophysics                                        | Expected, June 2029 |
| MS in Astronomy & Astrophysics                                         | June 2025           |
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| The University of Chicago                                              | Chicago, IL         |
| Bachelor of Arts in Physics, Bachelor of Science in Astronomy w/Honors | June 2023           |

## RESEARCH EXPERIENCE

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| Imprints of Dynamical Dark Energy in the Lyman-Alpha Forest<br>UC Santa Cruz: Dept. of A&A   Supervised by Brant Robertson                                                                                                                                                                                                                                                                                                                                      | September 2023 – Present  |
| <ul style="list-style-type: none"><li>Analyze high-resolution hydrodynamic cosmological simulations which incorporate a time-evolving dark energy equation of state</li><li>Highlight a spectral- and redshift- dependent feature in the transmitted flux power spectrum when comparing between dynamical dark energy and standard LCDM models</li></ul>                                                                                                        |                           |
| Chemo-Dynamic Properties of the Fastest Stars in the Milky Way<br>UT-Austin: Dept. of Astronomy   Supervised by Keith Hawkins                                                                                                                                                                                                                                                                                                                                   | June 2022 – October 2023  |
| <ul style="list-style-type: none"><li>Analyze chemical properties of fast stars in the Galaxy using Gaia DR3 to chemically tag them onto a stellar population</li><li>Analyze MCMC orbit integration of the fastest stars to physically determine their origins</li></ul>                                                                                                                                                                                       |                           |
| Caustic Properties in Outskirts of Dark Matter Halos<br>University of Chicago: Dept. of A&A   Supervised by Susmita Adhikari & Andrey Kravtsov                                                                                                                                                                                                                                                                                                                  | February 2021 – June 2023 |
| <ul style="list-style-type: none"><li>Analyze density histories of cold dark matter halos using cold dark matter simulations</li><li>Study phase space structures in solid angle bins to develop an algorithm to identify the splash back radius of cluster-sized dark matter halos in simulations</li></ul>                                                                                                                                                    |                           |
| LADEE Data Analysis of the Lunar Exosphere<br>NASA: Goddard Space Flight Center   Supervised by Orenthal J. Tucker                                                                                                                                                                                                                                                                                                                                              | June 2022 – August 2022   |
| <ul style="list-style-type: none"><li>Analyzed how chemical abundances in the Lunar Exosphere changes with respect to local lunar time</li></ul>                                                                                                                                                                                                                                                                                                                |                           |
| COOL-LAMPS<br>University of Chicago: Dept. of A&A   Supervised by Michael D. Gladders                                                                                                                                                                                                                                                                                                                                                                           | January 2022 – June 2022  |
| <ul style="list-style-type: none"><li>Studied strongly lensed galaxies as part of the “Chicago Optically-selected Lenses— Located at the Margin of Public Surveys” (COOL-LAMPS) collaboration</li><li>Identified galaxy candidates using extragalactic spectroscopic DECam Legacy Survey</li><li>Utilized the surface brightness modeling software GALFIT to describe galaxy mass distribution, providing information on stellar population synthesis</li></ul> |                           |

Study of Gaseous Argon Time Projection Chambers June 2021 – August 2021  
Fermi National Accelerator Laboratory | Supervised by David Caratelli & Angela Fava

- Analyzed setups of gaseous Argon time projection chambers for low energy nuclear recoils using simulation software of Argon ion trajectories into gaseous Argon to model, optimize, and study both position and energy resolution
- Created supplemental research material for electron charge amplification in liquid Argon

## PUBLICATIONS

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Mork, et. al., *COOL LAMPS VII: Quantifying Strong-lens Scaling Relations with 177 Cluster-scale Gravitational Lenses in DECaLS* ApJ (2025) doi: [10.3847/1538-4357/ada24c](https://doi.org/10.3847/1538-4357/ada24c)

Nelson, et. al., *A detailed chemical study of the extreme velocity stars in the galaxy* MNRAS (2024) doi: [10.1093/mnras/stae1656](https://doi.org/10.1093/mnras/stae1656)

## PRESENTATIONS

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*Exploring the Lyman-Alpha Forest with cosmological simulations in a dynamical dark energy cosmology* using Cholla

- UC Santa Cruz Friday Lunch Astrophysics Seminar Hour (FLASH), May 2025

*Dynamical Dark Energy Imprints in the Lyman-Alpha Forest*

- Arizona Winter School on Multi-Probe Cosmology in the Roman Space Telescope, January 2026
- Joint National Society of Black Physicist and National Society of Hispanic Physicists Conference, November 2025

*Chemo-Dynamic Properties of the Fastest Stars in the Milky Way using Gaia DR3*

- 241<sup>st</sup> Meeting of the American Astronomical Society, [January 2023](#)
- UT Austin Astronomy Department Summer Poster Session, August 2022

*Caustic Properties in the Outskirts of Dark Matter Halos*

- University of Chicago Undergraduate Research Symposium, May 2022
- Midstates Consortium for Math and Science, November 2021
- Society for the Advancement of Chicanos and Native Americans in Science (SACNAS) National Diversity in STEM, October 2021

## TEACHING EXPERIENCE

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Lecturer | Project for Inmate Education June 2024 – Present

- Teach astronomy, pre-algebra, and college algebra to incarcerated people at the Santa Cruz County Main Jail, allowing students to earn college credit at no personal cost

Teaching Assistant | Diverse Universe: Stars, Planets, Galaxies March 2025 – June 2025

- Develop supplementary material to lecture and lead weekly discussion sections for the “intro to astronomy” course aimed at students interested in pursuing an astronomy + astrophysics major
- Create homework solutions, grade homework, and be readily available to answer astronomy-related questions for the 11 enrolled students

Teaching Assistant | Diverse Universe: Stars, Planets, Galaxies March 2024 – June 2024

- Develop supplementary material to lecture and lead 2 weekly discussion sections for the “intro to astronomy” course aimed at students interested in pursuing an astronomy + astrophysics major
- Grade homework in a timely manner and be readily available to answer astronomy-related questions for the 30 enrolled students

Teaching Assistant | Computational Techniques in Astrophysics March 2022 – June 2023

- Hold open office hours 5-8 hours weekly helping students on coding projects related to techniques used in everyday astrophysics research including integration techniques, function approximation, Machine Learning, MCMC, and ODEs
- Spring Quarters of 2022 and 2023

Physics Tutor | Department of Physics October 2021 – March 2022

- Hold open hours for peer tutoring aimed at undergraduate students in the introductory physics sequences

### **SERVICE, OUTREACH, MENTORSHIP**

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UC Santa Cruz: Dept. of A&A: Graduate Admissions Committee December 2025 – Present

- Evaluate applicants to the Dept. of A&A’s graduate doctoral program on a holistic scale over their academic preparation, research experience, professional maturity, reaction to adversity, and community engagement
- Discuss nomination to shortlist, interview, and admit applicants with committee consisting of 5 faculty members and 4 graduate students

UC Santa Cruz: Dept. of A&A: DEI Committee October 2024 – Present

- Develop programming promoting diversity, equity, and inclusion for underrepresented peoples internal to the Dept. of A&A
- Schedule open discussions to highlight the available resources and roles of faculty members to support graduate students
- Organize panel discussions on the role of post-doctoral researchers and application to fellowships for graduate students
- Secure internal department funding for early career scientists to attend identity-based conference (2025 Joint National Society of Black Physicists and National Society of Hispanic Physicists Conference)

- UC Santa Cruz: Dept. of A&A: Colloquium Committee October 2024 – Present
- Gather suggested speakers from graduate students and post-doctoral researchers within the Dept. of A&A
  - Highlight and advocate for an invitation to traditionally underrepresented and early career scientists to present at weekly colloquium
  - Identify and communicate with graduate students who hold similar research interests as colloquium speaker to volunteer as graduate student host where they:
    - Present a Pre-Colloquium Paper Discussion the day before colloquium on a recently published paper by the colloquium speaker
    - Lead discussion during the graduate student lunch with the colloquium speaker
    - Guide speaker to lunch and either their guest office or next meeting
- Científico Latino: [Graduate Student Mentorship Initiative](#), Mentor August 2023 – April 2025
- Mentor two highly accomplished Latin undergraduate senior students through the graduate school application process for physics and astrophysics graduate programs
  - Provide timely feedback on essays and practice mock interview questions
- SEDS-UChicago Vice President, Treasurer March 2020 – June 2023
- Plan and coordinate weekly meetings, events, and workshops to help students with model rocketry and high-powered rocketry
  - Outreach and coordinate logistics to industry professionals to plan for the national conference SpaceVision 2022 hosted at UChicago
  - Manage the funds for my school's SEDS chapter. Write and defend grant proposals from SEDS-USA and UChicago Student Government's Financial Committee
- Moneythink Mentor November 2019 – June 2023
- Mentor local Chicago high school students on the importance and process of financial literacy. Provide advice on post-graduation plans including college and trade school
- SACNAS Secretary, Member October 2020 – June 2022
- Develop events and programming to engage with minority students on campus and take notes during leadership meetings
- Kavli Institute Space Explorers Instructor June 2021 – January 2022
- Design, develop, and document summer curriculum for high school students
- Odyssey Community Scholars Fellow September 2020 – June 2021
- Write, edit, and develop a First-Generation Low-Income Catalog documenting resources and providing advice from shared experiences at the University of Chicago for incoming students from disadvantaged backgrounds
  - Share important issues and suggested solutions with the Dean of Students in the University for the Odyssey Scholar community

## RELEVANT WORK EXPERIENCE

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Tech Staff / John Crerar: Computer Science Instructional Lab Tutor    October 2021 – June 2023

- Develop, maintain, and deploy software for over 50 Mac Pros, iMacs, and Mac minis
- Provide technology and hardware support for students

NASA L'SPACE Mission Concept Academy Member    January 2021 – May 2021

- Lead science-related proposals for a 55-page preliminary design review and a 30-minute presentation on a theorized mission to Venus's atmosphere in the 50-75 km altitude zone

Argonne National Lab Summer Instructor    June 2020 – September 2020

- Adapted the Education Department's Learning Center summer curriculum and programs to a virtual environment
- Developed curriculum design for the All About Energy educational program

## AWARDS & GRANTS

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|-----------------------------------------------------------------------------------------------|-------------|
| National Science Foundation: Graduate Research Fellowship Program                             | 2023 – 2028 |
| Osterbrock Graduate Leadership Program Fellow                                                 | 2025 – 2028 |
| Regents Fellowship, UC Santa Cruz                                                             | 2023        |
| Questbridge National College Match Scholar                                                    | 2019 – 2023 |
| <a href="#">Texas Astronomy Undergraduate Research for Under-represented Students Scholar</a> | 2022        |
| Department of Astronomy & Astrophysics Research Award                                         | 2022        |
| Jeff Metcalf Grant                                                                            | 2021        |
| Micro-Metcalf Grant                                                                           | 2020        |
| Hispanic Scholarship Fund Scholar                                                             | 2018        |

## OBSERVATION EXPERIENCE

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Public Data – Experience with data from Gaia and DECALS

Lick Observatory – Nickel Telescope, Shane KAST Spectrograph and ShARCS (5 Nights)

Magellan Telescope – IMACS Optical Imaging, FIRE/FourStar Near-Infrared (3 Nights)

McDonald Observatory – Cross-Dispersed Echelle Spectrograph (2 Nights)

## SKILLS

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|-------------|---------------------------------------------------------------------------------------------------|
| Programming | Python, C, C++, Bash, SQL/ADQL, Mathematica, LaTeX                                                |
| Astronomy   | Simulations, Data Analysis, Data Reduction                                                        |
| Software    | Microsoft Office, Adobe Photoshop, DS9, TOPCAT, GALFIT, Pulse Height Analyzer, Oscilloscope, SRIM |
| Languages   | English, Spanish                                                                                  |